Fatemehsadat Mousavinasab, Ph.D.

Atlanta, GA | fatemeh.kmls@gmail.com | +1 470-265-2561 | LinkedIn | Portfolio | Google Scholar
U.S. Permanent Resident (no sponsorship required)

RESEARCH SUMMARY

Immunologist studying how T lymphocytes develop, become tolerant, and go wrong in autoimmunity. My dissertation uses a new mouse model (HLA-DQ8 humanized, Foxp3-CNS1 mutant) to ask how peripheral regulatory T cells keep us tolerant to gluten in celiac disease. I also developed an improved $CD4^+$ hybridoma method, published in Bio-protocol (2026), to find autoreactive T-cell clones. Earlier, I worked on regulatory T cells and eosinophil markers in eosinophilic GI disease, and on macrophage polarization in IBD. I have 11 publications (6 research articles, 4 reviews, 1 book chapter).

EDUCATION

Ph.D., Biomedical Sciences, Georgia State University, Atlanta, GA *Expected Dec 2026 – Mar 2027*
GPA 4.19, ranked first in class. Dissertation: Peripheral regulatory T cells in oral tolerance to gluten in celiac disease. Mentor: Leszek Ignatowicz, Ph.D.

M.Sc., Medical Immunology, Shahid Beheshti University of Medical Sciences, Tehran, Iran *2013 – 2016*
GPA 3.86. Thesis: Regulatory T cells in esophageal tissue of patients with eosinophilic esophagitis, compared with GERD and controls.

B.Sc., Medical Laboratory Sciences, Arak University of Medical Sciences, Iran *2008 – 2012*
GPA 3.93

RESEARCH EXPERIENCE

Ph.D. Candidate, Center for Translational Immunology, Georgia State University *2022 – Present*

- Built a new mouse model of celiac disease (HLA-DQ8 humanized, Foxp3-CNS1 mutant, so it lacks peripheral regulatory T cells, with a Foxp3-GFP reporter) to test how these cells maintain oral tolerance to gluten.
- Designed and ran in vivo studies with four groups (CNS wild-type vs. mutant, gluten-free vs. regular diet), plus adoptive transfer and cholera toxin models.
- Profiled T cells from the intestine and lamina propria (Th17/Treg balance, FOXP3, RORγt, cytokines) using multicolor flow cytometry, histology, and functional assays.
- Set up Nur77-GFP $CD4^+$ T-cell hybridomas specific for human class II MHC/peptide complexes (thymoma fusion, HLA-DQ8 dendritic cell co-culture, peptide testing). With them I found more autoreactive $CD4^+Foxp3^-$ clones in humanized mice.
- Co-authored a step-by-step protocol (Bio-protocol, 2026) so other labs can repeat the hybridoma method. Manage mouse colonies and do injections, gavage, blood sampling, surgery, and organ isolation.

Head of Immunology & Instructor, GenIran Research Lab, Tehran, Iran *2020 – 2022*

- Led the immunology section and research projects that used immunoassays. Managed quality control for ISO 17025.

Researcher, Research Institute for Gastroenterology & Liver Diseases, Tehran, Iran *May – Dec 2019*

- Designed IBD immunology projects, including one on macrophage polarization during acute inflammation and after treatment.

M.Sc. Researcher, Shahid Beheshti University of Medical Sciences, Tehran, Iran *2013 – 2016*

- Counted regulatory T cells in esophageal tissue from patients with eosinophilic esophagitis, GERD, and controls. Published in Allergologia et Immunopathologia.

PUBLICATIONS (11)

- **Mousavinasab F**, Szurek EA, Cebula A, Ignatowicz L, Kuczma MP. Improved protocol for establishing CD4+ hybridomas specific for human class II MHC/peptide complex. *Bio-protocol*. 2026;16(13):e5737.
- Khatami S, Ebrahimi S, Mahjoub FE, et al. (incl. **Mousavinasab F**). Eosinophilic cell-free granules (CCR3, MBP) in colonic biopsy of pediatric patients with eosinophilic colitis. *Iran J Immunol*. 2026;23(2).
- Ebrahimi S, Khatami S, Mahjoub FE, et al. (incl. **Mousavinasab FS**). Eosinophilic cell-free granules (CCR3, MBP) in esophageal biopsy of patients with eosinophilic esophagitis. *Pediatr Gastroenterol Hepatol Nutr*. 2025;28(1):19-26.
- **Mousavinasab F**, Babaie D, Nilipour Y, et al. More regulatory T cells in esophageal tissue of patients with eosinophilic esophagitis than in GERD and controls. *Allergol Immunopathol*. 2019;47(5):431-436.
- Abdolahy M, Rasouli Sh, Babaie D, et al. (incl. **Mousavinasab F**). More regulatory T cells in peripheral blood of children with eosinophilic esophagitis. *Gastroenterol Hepatol Bed Bench*. 2021:25-30.
- Al-Zaidi HMH, **Mousavinasab F**, Radseresht N, et al. GJB2 and SLC26A4 genes in patients with Pendred syndrome deafness. *Cell Mol Biomed Rep*. 2023;3(3):163-171.
- **Reviews and book chapter:** Aghaei-Zarch SM et al., *Biomed Pharmacother* 2023 (particulate matter and apoptosis). Hoorzad P et al., *Diabetes Res Clin Pract* 2023 (lncRNA/miRNA-NFκB network in diabetes). **Mousavinasab F** et al., *Cancer Cell Int* 2023 (microbiome modulation in inflammatory diseases). **Mousavinasab F**, Razani E, *Iran J Blood Cancer* 2023 (vesicular stomatitis virus for cancer therapy). Salehi-Sardoei A et al., Elsevier 2024, Ch. 11 (endophytic fungi).

- Full list with links to every paper: Google Scholar.

PRESENTATIONS

- Digestive Disease Research Day, Emory–Georgia State University (Sep 2026): Peripheral regulatory T cells in oral tolerance to gluten, through a new mouse model of celiac disease. Dr. Suresh Sitaraman Award, top three abstracts.
- AAI Immunology 2026, Boston, MA (Apr 2026): Peripheral regulatory T cells in oral tolerance to gluten in celiac disease. Poster Presentation Award.
- 4th Annual Graduate Conference for Research, Scholarship, and Creative Activity, Atlanta, GA (Feb 2026): More autoreactive $CD4^{+}Foxp3^{-}$ T-cell clones in humanized HLA-DQ8/$Ab^{-/-}$ mice, found with Nur77-GFP hybridomas.
- Same mouse model presented at the 2025 Southeast Regional Clinical & Translational Science Conference (Pine Mountain, GA), the 3rd Annual Graduate Conference (Feb 2025), and the 2nd Research Day, Institute for Biomedical Sciences.

HONORS & AWARDS

- Dr. Suresh Sitaraman Award (top three abstracts), Digestive Disease Research Day, Emory–Georgia State University (2026). Poster Presentation Award, AAI Immunology 2026.
- Best Oral Presentation, Basic and Advanced Translational Immunology Course, Jaipur, India, and FIMSA travel grant (2019). EFIS travel grant, SEEIS 2018, Yerevan. EAACI junior member travel grant, Helsinki (2017).

TECHNICAL SKILLS

Immunology & cell biology: Multicolor flow cytometry (FlowJo), FACS/MACS sorting, primary cell culture, dendritic cell generation, T-cell hybridomas and peptide testing, Treg suppression assays

Molecular biology & protein: DNA/RNA isolation, PCR, qPCR, ARMS-PCR, RFLP, primer design, cDNA synthesis, ELISA, Western blot, SDS-PAGE, BCA, exosome isolation

Histology & in vivo: H&E, immunohistochemistry, immunofluorescence, ImageJ; mouse colony management, disease induction (celiac, colitis), injections, gavage, surgery, organ isolation

Data & software: GraphPad Prism, SPSS, REST, FlowJo, ImageJ, EndNote; NCBI/BLAST, genome and protein databases

TEACHING, MENTORING & SERVICE

- Teaching assistant (2017–2019): taught practical immunology to master's students and advised their projects. Started an immunoassay course and taught a molecular genetics course at GenIran (2020–2022).
- Vice President, Translational Biomedical Sciences Student Association, Georgia State. Judge, Georgia State Undergraduate Research Conference (2024, 2025). Peer reviewer, Cellular, Molecular and Biomedical Reports (2023, six manuscripts).
- Member: AAI, IUIS, EAACI, FIMSA (Women Speakers).

REFERENCES

Leszek Ignatowicz, Ph.D., Distinguished University Professor, Center for Translational Immunology, Georgia State University | lignatowicz@gsu.edu | 404-413-6685

Cynthia Nau Cornelissen, Ph.D., Distinguished University Professor; Director, Center for Translational Immunology, Georgia State University | ccornelissen@gsu.edu | 404-413-6680